

[143] As regular electrocardiogram is relatively insensitive, there should be a low threshold to use Holter monitoring.

Renal involvement

Acute kidney injury associated with microangiopathic haemolytic anaemia (MAHA) and accelerated phase hypertension is a serious and potentially fatal complication of systemic sclerosis (scleroderma renal crisis; SRC). It most commonly occurs in patients with the progressive, diffuse form of the disease and within the first 4 years of disease onset. The presence of anti-RNA polymerase III antibodies is a significant risk factor and is detected in approximately one-third of SRC cases.[144]

Therefore, regular blood pressure monitoring (at least twice weekly, preferably via home monitoring) is recommended to enable early detection of renal involvement. Glucocorticoid use at doses exceeding 15 mg of prednisone equivalents per day should be avoided, as they are associated with an increased risk of renal crisis and have long-term adverse effects.[144]

A small subset of patients may develop normotensive renal crisis, often presenting with features of thrombotic microangiopathy. Chronic renal involvement in systemic sclerosis is characterized by a slowly progressive obliterative vasculopathy. Urinary protein excretion has been identified in multiple studies as a major independent risk factor for mortality.[145] Hence, annual assessment of urinary protein is advised.

Consensus criteria for the classification and diagnosis of SRC are currently under development and validation.[146] Additionally, there is a need for improved management of chronic kidney disease (CKD) in systemic sclerosis, along with the development of more effective biomarkers and therapeutic strategies.

General recommendation for a regular diagnostic work-up in patients with SSc

Following an initial baseline assessment (see Table 6), lifelong follow-up with at least annual evaluations is recommended due to the chronic and potentially progressive nature of systemic sclerosis. Patients with active or progressive disease should be monitored more frequently.

The annual evaluation should include a comprehensive clinical assessment, including modified Rodnan Skin Score (mRSS), and the following:

- Pulmonary function tests with plethysmography and measurement of diffusing capacity for carbon monoxide (DLCO)
- Blood pressure measurement
- Electrocardiogram
- Echocardiography
- Erythrocyte sedimentation rate (ESR) / C-reactive protein (CRP)
- Complete blood count
- Clinical chemistry (liver function tests, creatinine, urea)
- Urinary protein assessment

In patients at increased risk for renal crisis—such as those with progressive diffuse disease or anti-RNA polymerase III antibodies—frequent blood pressure monitoring is strongly recommended, preferably through home-based measurement (see Table 7).

Treatment

Therapy for skin involvement

Treatment of Raynaud's phenomenon

Avoidance of cold exposure and consistent protection from cold are essential. Heated gloves, shoes, and hand warmers are commonly used. Additional

supportive measures include paraffin baths, heated seed pillows, therapy balls, and physical therapy.